

Manipulation of Deformable objects using VLM and Taxonomy for Long Horizon Tasks

Gawtam Chithra Ramesh*, David Blanco-Mulero[†], Yifei Dong*

*KTH Royal Institute of Technology, Sweden

[†]Universitat Politècnica de Catalunya (UPC), Spain

Abstract—This work addresses the challenging problem of manipulating deformable objects (DOMs) by exploring the integration of Vision-Language Models (VLMs) with structured knowledge from taxonomies. While VLMs demonstrate strong capabilities in high-level reasoning and scene understanding, their application to DOM manipulation necessitates a deeper understanding of deformation modes and interaction dynamics. To bridge this gap, we leverage the recently introduced T-DOM taxonomy, which provides a comprehensive classification of deformable object behaviors and relevant robot actions. Our work investigates the potential of combining the perceptual and reasoning abilities of a state-of-the-art VLM with the structured knowledge of T-DOM to enable more effective long-horizon planning for DOM manipulation tasks. Through simulation experiments, we evaluate the performance of this integrated approach, identify key challenges and opportunities, and lay the groundwork for future research in real-world deformable object manipulation.

Index Terms—VLM, Taxonomy, Deformable Object Manipulation

I. INTRODUCTION

The field of robotic manipulation has witnessed significant progress, driven by the increasing demand for automation across various sectors. However, while robots excel at handling rigid objects, the manipulation of deformable objects (DOMs) remains a significant challenge. Unlike rigid bodies with well-defined states, DOMs possess infinite-dimensional state spaces and exhibit complex, often unpredictable, dynamic behaviors, as highlighted in prior work. This inherent complexity makes fundamental tasks such as grasping, transporting, and assembling far more intricate. The limitations of traditional force and tactile sensing in dealing with the varying responses of deformable materials further underscore the need for advanced approaches capable of effectively perceiving, understanding, and interacting with these entities.

A. Background and Motivation

Vision-Language Models (VLMs) have recently demonstrated unprecedented capabilities in areas such as logical reasoning, commonsense inference, and visual scene understanding [3], [7]. By unifying perceptual inputs with language-based reasoning, VLMs offer a promising avenue for tackling the complexities of robotic manipulation, particularly in scenarios demanding intricate perception and high-level reasoning. These models enable robots to generate coherent high-level plans grounded in visual data [5], [8], opening new possibilities for robotic manipulation where long-horizon

planning and semantic understanding are critical. Recent work, such as RoboDexVLM [6], highlights the potential of VLMs for dexterous manipulation. Robotic grasping and manipulation inherently involve challenges ranging from detailed scene understanding to precise contact-rich interactions. To navigate this complexity, taxonomies have long served as a systematic framework, guiding both end-effector design and algorithm development [2], [4]. However, a significant limitation of most existing taxonomies is their assumption of rigid bodies, largely overlooking the complexities introduced by deformation phenomena.

B. Current Challenges

Despite the advancements in VLMs and the established utility of taxonomies in robotics, the effective integration of these two powerful paradigms for the manipulation of deformable objects remains largely unexplored. Existing robotic manipulation taxonomies primarily focus on rigid bodies and do not adequately address the unique challenges posed by deformation. While VLMs offer rich perceptual understanding and reasoning, they often lack the structured knowledge and explicit representation of deformation modes necessary for precise manipulation planning. This gap necessitates the development of novel approaches that can bridge the semantic understanding of VLMs with a structured understanding of how deformable objects behave and can be manipulated. The varying response of deformable objects to traditional sensing methods further complicates the problem, requiring solutions that go beyond conventional force and tactile feedback. Benchmarking efforts like ManipBench [9] are crucial for evaluating the performance of VLMs in low-level manipulation tasks.

C. Contribution

To address the aforementioned challenges, this work investigates the integration of a state-of-the-art VLM, specifically GPT-4o, with a taxonomy for the manipulation of deformable objects, building upon the recently introduced T-DOM (Fig. 1), which provides a detailed classification of object deformations alongside robot motion and interaction modalities. The primary contributions of this research include:

- An evaluation of the performance of a state-of-the-art VLM when used in conjunction with a taxonomy for the manipulation of deformable objects, providing insights into the strengths and weaknesses of this integrated approach.

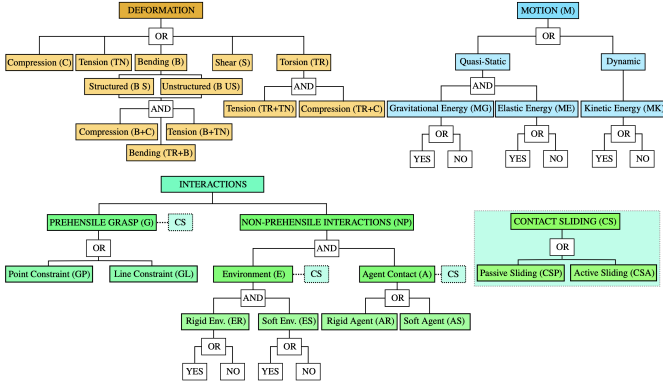


Fig. 1. Taxonomy for Deformation, Motion and Interaction proposed by Blanco-Mulero et al. in [1]

- The identification of key challenges and opportunities associated with employing VLMs for understanding and planning tasks involving deformable objects, with a particular focus on how a taxonomy can address the inherent complexities of deformation.
- The proposal of a novel framework or pipeline that tightly integrates a VLM with a taxonomy to facilitate long-horizon manipulation of deformable objects, likely incorporating a hierarchical reasoning approach.
- Experimental results, primarily obtained through simulations in PyBullet, demonstrating the feasibility and effectiveness of the proposed approach in tackling a range of deformable object manipulation tasks, with preliminary steps towards real-world validation to assess the transferability of the simulated results.

D. Report Structure

The remainder of this report is structured as follows: Section II details the preliminary experiments conducted, including the setup, results, and lessons learned. Section IV outlines the proposed methodology for the final experiments, elaborating on the enhanced integration of VLMs and taxonomies, improved scene understanding techniques, refined low-level planning strategies, and the design of the experimental tasks. Section V and Section VI describes the evaluation metrics and key measurements that will be used to assess the performance of the proposed approach. Section VII discusses the expected outcomes of the final experiments and provides a broader discussion of the research. Finally, Section VIII concludes the report with a summary of the work and potential directions for future research.

II. INITIAL EXPERIMENTS AND RESULTS

The initial phase of this research involved setting up a preliminary experimental framework to explore the feasibility of using Vision-Language Models (VLMs) for interpreting and planning deformable object manipulation tasks. The primary software used during this phase included the gptv4o GUI for prompt design and preliminary evaluation. The OpenAI GUI was also utilized for initial assessments of taxonomy

code generation for individual frames. The specific prompts provided to ChatGPT to interpret the taxonomy were designed to instruct the model on the structure and meaning of the taxonomy codes. For instance, an instruction such as "Understand the taxonomy tree" was given, followed by a sample taxonomy code representing a robot arm approaching a towel. An example from the preliminary test is available in the Appendix A.

Initial tests assessed the VLM's capacity to generate appropriate taxonomy codes from visual inputs, typically single frames or start-and-end frame pairs depicting actions such as grasping a towel or manipulating a glove. The objective was to gauge the VLM's foundational understanding of these manipulation tasks and its ability to map visual information to the structured representation offered by the taxonomy. These early evaluations indicated that while the VLM could identify high-level subgoals and corresponding taxonomy codes given start/end frames and a task prompt, it struggled with more specific actions, like passive sliding, when provided with generic prompts. Performance in identifying primitives improved with more detailed prompting.

Given the noise in real-world image data used initially, and to facilitate more controlled experimentation, the research transitioned to the PyBullet simulation environment. This shift was primarily motivated by PyBullet's robust capabilities for simulating soft body dynamics, essential for accurately modeling deformable objects (prompts used with PyBullet are detailed in Appendix A).

To integrate the VLM with the simulation, a pipeline was developed (illustrated in Figure 1, which corresponds to the conceptual diagram vlm-taxonomy.drawio.pdf). Within this pipeline, the VLM API (Gemini) is tasked with interpreting the scene rendered by PyBullet and formulating a high-level plan to achieve the specified manipulation. This plan is subsequently processed by a low-level planner, which translates the VLM's abstract instructions into a concrete end-effector trajectory for the simulated robot. While the pipeline was successfully constructed, its performance was not extensively validated due to observed inaccuracies in the VLM's scene interpretation and challenges in consistently translating correct high-level plans into successful low-level actions. Initial work also considered aspects such as video segmentation for scene understanding.

Preliminary quantitative results were obtained by evaluating the VLM's scene interpretation and taxonomy-code generation. For example, the VLM was given a task to interpret the two states in a towel folding task. These states consisted of 128x128 pixel images, showing a robotic arm interacting with a towel (Figure 3). Table I summarizes the F1-Score and BERT similarity metrics for taxonomy classification for these example states. Further details on the exact human-labels and VLM-generated labels are provided in Appendix C. These initial findings and the challenges encountered informed the subsequent analysis and the design of the final experiments.

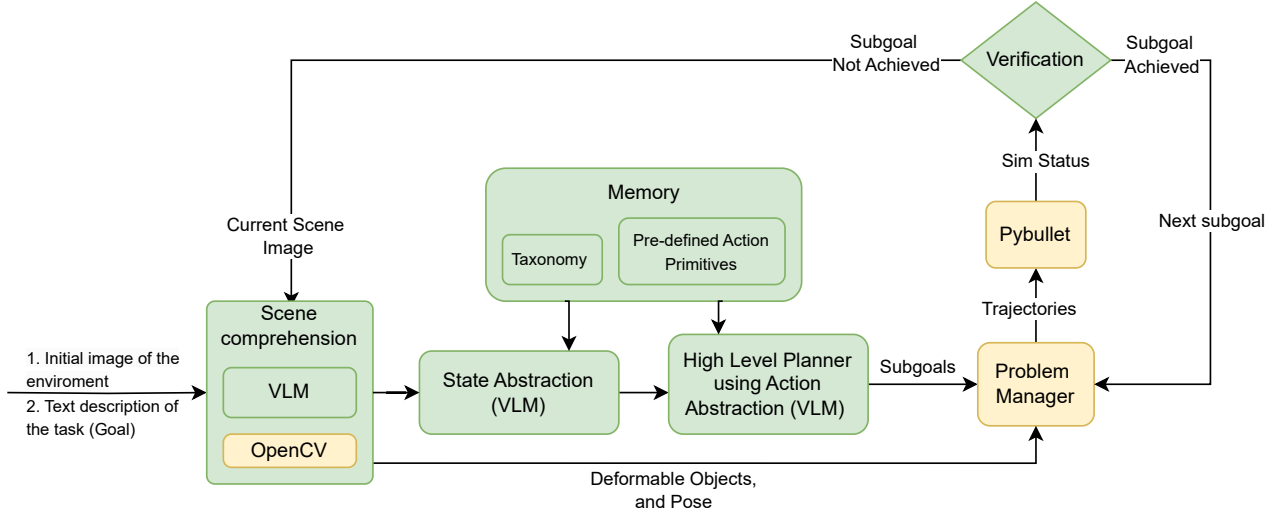


Fig. 2. Proposed pipeline for the experiment setup using Pybullet and VLM (gemini)

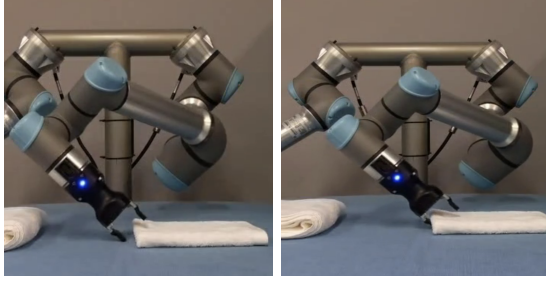


Fig. 3. Example of a robot approaching the towel to pick it up

TABLE I
PERFORMANCE METRICS FOR TAXONOMY CLASSIFICATION FOR THE
EXAMPLE IN FIG. 3

	F1-Score	BERT
state_1	0.8	0.7594
state_2	1.0	0.7513

III. 3. ANALYSIS OF RESULTS

Preliminary experiments revealed key insights into applying Vision-Language Models (VLMs) to deformable object manipulation. A primary observation was the VLMs’ difficulty in accurately interpreting object deformation, contrasting with their stronger performance in understanding motion and interaction. This suggests potential limitations in current VLMs’ capacity to capture the nuanced visual cues and complex dynamics of shape changes in deformable objects. Such limitations could stem from training data lacking sufficient coverage of deformation phenomena or from model architectures not being optimized for extracting these specific visual features.

Conversely, the VLMs’ proficiency in comprehending motion and interaction highlights their potential for high-level planning and control in robotic manipulation. The ability to understand action sequences and robot-environment interplay

is crucial for many manipulation tasks, underscoring the VLM’s utility in this domain.

The experiments also emphasized the inherent challenges in applying VLMs to deformable objects. These objects exhibit high degrees of freedom and often unpredictable behavior, complicating generalization for VLMs from limited examples. The complex dynamics and infinite-dimensional state space of deformable objects present a significant hurdle for models predominantly trained on static images and rigid object interactions.

A critical refinement in the experimental methodology was the transition to the PyBullet simulation environment. PyBullet’s specialized soft body simulation capabilities facilitated a more controlled and realistic representation of deformable object behavior, enabling a focused analysis of VLM performance in this area. This underscores the importance of selecting appropriate simulation environments tailored to specific task requirements.

Finally, the initial experiments highlighted the necessity of meticulous prompt design for VLMs. The phrasing used to describe tasks and scenes significantly influences the VLM’s interpretation and the accuracy of its outputs, such as generated taxonomy codes. The potential for more direct integration of structured knowledge, like the established taxonomy, into the VLM’s processing also emerged as a promising avenue for future work. This could involve utilizing the taxonomy not merely to guide prompts but also as an output framework, potentially enhancing the reliability and accuracy of VLM responses.

IV. FINAL EXPERIMENTS

The subsequent research will focus on enhancing the capability of general-purpose VLMs to interpret scenes involving deformable object manipulation (DOM), particularly concerning how textual outputs can be translated into continuous ac-

tions or trajectories. Recognizing this as a complex area, a significant portion of the effort will be dedicated to analyzing this translation process, which is considered a valuable contribution in itself. The primary aim is to leverage a predefined taxonomy to guide VLMs in understanding deformations, interactions (gripper-object, object-environment), and the intricate motions involved in DOM.

The final experiments are designed with a twofold objective:

- To assess and enhance VLM performance through integration with the taxonomy.
- To explore practical applications by incorporating the VLM-taxonomy system into a comprehensive pipeline. This will involve evaluating system performance in a continuous simulation environment using PyBullet and, if feasible, testing with real robots to assess sim-to-real transferability.

The planned approach encompasses several key areas:

Improving VLM Integration with Taxonomy: Exploration will move beyond using the taxonomy solely for prompt guidance towards a more profound integration. This may involve the taxonomy functioning as a structured output constraint for the VLM, potentially through fine-tuning the VLM to directly predict taxonomy codes or by using the taxonomy to filter and rank VLM outputs.

Improved Scene Understanding: Techniques to augment the VLM’s spatial reasoning will be incorporated. This could include utilizing depth information as an additional input modality for the VLM, alongside standard RGB images.

Experiment Design and Tasks: A curated set of manipulation tasks with progressively increasing complexity will be employed. Initial tasks will involve basic actions such as unfolding cloth or manipulating a rope, advancing to more demanding tasks like shaping clay or assembling deformable components. Variations within each task—such as different initial object configurations, robot starting positions, or the presence of obstacles—will be introduced to evaluate the VLM’s generalization capabilities.

Complete Pipeline Implementation: A hierarchical reasoning framework is envisioned (as depicted in the provided conceptual pipeline in Figure 2). The VLM will first generate a high-level plan in natural language. This plan will then be mapped to a sequence of taxonomy codes, which, in turn, will retrieve predefined motion primitives or guide a low-level planner. This strategy leverages the VLM for abstract reasoning and the taxonomy for specifying concrete actions.

Simulation and Real-World Validation: PyBullet will continue to be the primary platform for efficient experimentation and data generation, ensuring the simulation is calibrated to mirror real-world physics as closely as possible. A final stage of real-world validation is planned to assess the approach’s transferability, potentially using a robot arm and simple deformable objects like cloths or sponges.

Benchmarks: Experimental results will be compared against established DOM benchmarks such as SoftGym, PlasticineLab, ManipBench, or PhysBench to contextualize performance.

Refined Low-Level Planning: Consideration will be given to employing a dynamics-aware planner that accounts for the deformable object’s properties, potentially by incorporating a simplified deformable object model into the planning process, instead of a basic low-level planner. Furthermore, reinforcement learning techniques will be investigated for training the robot to execute motion primitives based on taxonomy codes, enabling adaptation to the inherent variability of deformable object manipulation.

V. KEY MEASUREMENT

Taxonomy codes, Textual correctness of the interpretation

Thoroughly analyse what aspects of DOM are easy to interpret by a VLM, what is very tricky (why is it tricky, how taxonomy helps it) explore ways to make it better.

The performance of the proposed approach will be evaluated using a range of quantitative metrics to assess different aspects of the system’s capabilities.

Taxonomy Code Accuracy: The accuracy of the VLM in predicting the correct taxonomy code for a given state or sequence of actions will be computed. This will involve using metrics such as exact match accuracy, which measures whether the entire predicted code is identical to the ground truth code, and partial match accuracy, which assesses the correctness of individual elements within the code. Additionally, the Hamming distance between the VLM-predicted code and the ground truth code will be calculated. This metric quantifies the number of substitutions, insertions, or deletions required to transform one code into the other, providing a measure of the dissimilarity between the predicted and actual taxonomy representations.

Textual Correctness of Interpretation: Natural Language Processing (NLP) metrics will be employed to evaluate the VLM’s textual description of the scene and the actions being performed. Specifically, BLEU (Bilingual Evaluation Understudy), ROUGE (Recall-Oriented Understudy for Gisting Evaluation), and METEOR (Metric for Evaluation of Translation with Explicit Ordering) scores will be calculated to compare the VLM’s generated description to a human-generated ground truth description. Furthermore, the BERT score will be used to measure the semantic similarity between the VLM’s description and the ground truth. As mentioned in the preliminary work, this can involve embedding the human description of the task using CLIP and then embedding the prompt given by the LLM, comparing the distance between these embeddings in the latent space.

Analysis of DOM Aspects: To gain a deeper understanding of the system’s performance across different facets of deformable object manipulation, the tasks and sub-actions will be categorized based on the T-DOM taxonomy (e.g., deformation type, motion type, interaction type). For each of these categories, the taxonomy code accuracy and textual correctness metrics will be calculated. This will allow for a quantitative assessment of which aspects of DOM the VLM handles well and which pose greater challenges. Confusion matrices will also be analyzed to identify specific types of

errors the VLM makes, such as confusing different types of deformation (e.g., tension with bending) or misclassifying contact types.

Robot Task Performance: The success rate of the robot in completing the defined manipulation tasks will be measured. This could be a binary metric (success/failure) based on whether the task’s goal state is achieved, or a more granular metric, such as the percentage of the task that has been completed. The efficiency of the robot’s actions will be quantified by measuring the time taken to complete a task, the total number of steps required, or the length of the robot’s trajectory. Additionally, the accuracy of the robot’s actions will be assessed by measuring the precision with which it reaches target locations or manipulates the deformable object to achieve the desired configuration.

Motion Primitive Prediction: If the methodology incorporates the use of motion primitives, the accuracy of the VLM in predicting the correct sequence of motion primitives for a given task will be measured. Furthermore, the smoothness and feasibility of the robot trajectories generated from the predicted motion primitives will be evaluated to ensure that the planned motions are physically realizable and do not lead to jerky or unstable behavior.

Qualitative Evaluation In addition to the quantitative metrics, a qualitative evaluation will be conducted to provide further insights into the system’s performance and limitations. This will involve including illustrative examples of both successful and failed manipulation sequences, accompanied by the corresponding VLM descriptions and taxonomy codes. Visualizations, such as highlighting regions of interest in the input images or plotting the robot’s trajectories in the simulation environment, will be used to provide a more intuitive understanding of the system’s behavior and to identify potential areas for improvement.

VI. EVALUATION

A. Quantitative Metrics

Calculate descriptive statistics (mean, standard deviation, min, max) for all quantitative metrics (taxonomy code accuracy, textual correctness scores, robot task performance metrics, motion primitive prediction accuracy, computational efficiency). Present these statistics in tables and graphs to provide a clear overview of the system’s performance. Include error bars in graphs to show the variability of the results.

B. Impact of Taxonomy

Conduct a comparative analysis to evaluate the contribution of the taxonomy, by comparing the performance of the VLM with and without the taxonomy guidance. Analyze how the taxonomy affects the VLM’s ability to handle different aspects of deformable object manipulation. For example, does the taxonomy improve the accuracy of deformation classification more than motion classification?

C. Generalization Ability

Evaluate the VLM’s generalization performance by testing it on new deformable object manipulation tasks or variations of existing tasks that were not included in the training set. Quantify the drop in performance on these unseen tasks compared to the training tasks. This will provide a measure of how well the VLM can generalize.

D. Error Analysis

Perform a detailed error analysis to understand the types of mistakes the VLM and the robot make. Categorize errors in taxonomy code prediction (e.g., misclassifying deformation types, incorrect action primitives). Analyze failure cases in robot task execution (e.g., failure to grasp, inaccurate manipulation). Identify patterns in the errors and discuss potential reasons for them (e.g., limitations of the VLM, deficiencies in the training data, challenges in the simulation).

E. Qualitative Evaluation

Include qualitative examples to illustrate the system’s performance and limitations. Show successful manipulation sequences with corresponding VLM descriptions and taxonomy codes. Present failure cases and discuss the reasons for the failure.

VII. EXPECTED OUTCOME

The primary objective of this work is to demonstrate the feasibility and effectiveness of utilizing a structured taxonomy to guide Vision-Language Models (VLMs) in interpreting and planning deformable object manipulation tasks. It is anticipated that by incorporating the taxonomy, the VLM will achieve a more structured and accurate understanding of the complex dynamics involved in manipulating deformable objects. A successful outcome would be characterized by a significant improvement in the VLM’s performance in taxonomy code prediction and task execution when compared to a baseline approach that does not utilize the taxonomy for guidance.

Quantitatively, this could be represented by:

- Achieving an average taxonomy code accuracy of 0.8 or above across the tested manipulation tasks.
- Demonstrating a statistically significant improvement in textual correctness metrics (BLEU, ROUGE, METEOR, BERT) when the VLM is guided by the taxonomy compared to when it is not.
- Achieving a high success rate (e.g., above 80%) in the robot’s execution of the manipulation tasks in both simulation and, if feasible, real-world scenarios.

A. Discussion

Given the inherent complexities associated with deformable object manipulation and the current limitations of Vision-Language Models, realistic expectations for the final outcome include demonstrating a clear improvement in VLM performance when guided by the taxonomy, although achieving perfect accuracy across all tasks may not be feasible. It is

also anticipated that the research will identify specific types of deformable object manipulation where the proposed approach is most effective, as well as areas where further research and development are needed to overcome remaining challenges. Ultimately, this work aims to provide a set of guidelines and best practices for effectively using VLMs in conjunction with taxonomies for the manipulation of deformable objects, contributing valuable insights to this area of research.

VIII. CONCLUSION

The research undertaken thus far has provided a foundational understanding of the challenges and opportunities associated with using VLMs for the manipulation of deformable objects. Preliminary experiments highlighted the VLM’s relative strength in interpreting motion and interaction compared to its difficulty in accurately understanding deformation. This underscored the need for strategies to enhance the VLM’s comprehension of deformable object dynamics, potentially through the integration of more structured knowledge. The proposed methodology for the final experiments focuses on a tighter integration between the VLM and a taxonomy for deformable object manipulation, employing a hierarchical reasoning approach. The evaluation plan includes a comprehensive set of quantitative and qualitative metrics to assess the performance of the proposed system. The expected outcomes point towards a significant improvement in the VLM’s ability to interpret and plan DOM tasks with the aid of a taxonomy.

A. Future Work

Building upon the findings of this research, several promising directions for future work can be explored. One avenue involves investigating the performance of different VLMs and their underlying architectures on the proposed DOM tasks, to determine which models are best suited for this challenging domain. Exploring alternative taxonomies or developing novel methods for structured knowledge representation that are specifically tailored for the nuances of deformable object manipulation could also yield valuable advancements. Expanding the range and complexity of the manipulation tasks to include more intricate scenarios and a wider variety of deformable object types would provide a more comprehensive evaluation of the proposed methodology’s generalizability. Conducting more extensive real-world validation experiments with different robotic platforms and deformable objects is crucial for assessing the practical applicability of the research. Investigating the integration of additional sensory modalities, such as haptic feedback, could potentially provide robots with a richer understanding of their interactions with deformable objects, leading to improved manipulation performance. Finally, exploring the application of the proposed approach to specific real-world problems in areas like textile manipulation for the garment industry, food handling in culinary applications, or surgical assistance in healthcare could demonstrate the practical impact and potential of this research.

B. Targeted Venue

Targeting IROS2026 for conference publication and a full version with detailed descriptions for IJRR.

ACKNOWLEDGMENT

I thank Prof. Florian T. Pokorný for providing valuable feedback. And the division of RPL at KTH for providing a great environment. I used DeepMind’s Gemini for language editing and proofreading, with all content and interpretations provided by the authors.

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APPENDIX

A. Scene Interpretation using gptv4o GUI

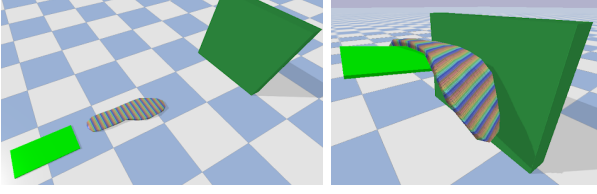


Fig. 4. The start and end goal states for the task of scooping an insole in pybullet environment

TABLE II
VLM’S HIGH LEVEL PLAN FOR TASK TO LIFT INSOLE

Subgoal	State Changes	Actions Required
1. Contact Established	Flat object touches deformable object, no deformation	Quasi-static motion, no sliding
2. Push to Wall	Compression & bending increase, wall interaction active	Quasi-static motion, passive sliding
3. Scooping Begins	Contact sliding starts, shear forces present	Dynamic motion, active sliding
4. Object Lifted	Less contact with the wall, possible shear	Dynamic motion, lifting

TABLE III
VLM’S HIGH LEVEL PLAN FOR TASK TO LIFT AN INSOLE AND PLACE IN A BOWL

Subgoal	State Changes	Actions Required
1. Contact Established	The end effector makes contact with the insole, no deformation yet.	Quasi-static motion, no sliding.
2. Push to Wall	Compression and bending increase as the insole is pressed. Wall interaction starts.	Quasi-static motion, passive sliding.
3. Sliding Up	Contact sliding starts; shear forces help lift the insole.	Dynamic motion, controlled sliding.
4. Detachment Begins	The lower part of the insole loses ground contact, lifting occurs.	Dynamic motion, lifting forces applied.
5. Reorientation	The insole is fully lifted, stabilized in midair for placement.	Controlled motion to align for placement.
6. Placement in Bowl	The insole is placed inside the bowl with minimal deformation.	Quasi-static motion, careful release.

B. Scene interpretation using gemini API

Obtained command (or 'exit' to quit) scoop the fish
Starting scene Comprehension ...
Identifying Taxonomy of the object in the scene
Waiting for response
json

```
{
  "deformable type": "Bending",
  "deformable axis": "x-axis",
  "interaction type": "Non-prehensile",
  "interaction with": "Environment",
  "interaction nature": "Rigid"
}
```

The fish appears slightly bent or flattened as it rests on the ground due to gravity. This can be categorized as bending along the approximate vertical or x-axis from our viewpoint/camera perspective. The fish is in contact with the ground, which is a non-prehensile interactive with the environment, and this interaction is rigid as it is simply resting under its own weight.
Creating a high level plan:

Plan:
1. Move the shovel towards the fish from its current position to get closer to the fish.
2. Lower the shovel vertically downwards such that the scoop part of the shovel is positioned just below the fish.
3. Move the shovel forward in the direction of the fish’s length, to slide the scoop underneath the fish and scoop it up.
4. Lift the shovel vertically upwards to raise the fish, now resting in the scoop of the shovel.

Listing 1. Terminal output showing scene comprehension and planning.

C. Comparisons of VLM Labels with Human Label

```
{
  "state_1": {
    "code": "010-00101011-0000",
    "reason": "The arm is in position to move under the towel, with no grasp or contact with the environment. The towel is not deformed."
  },
  "state_2": {
    "code": "010-00111011-0000",
    "reason": "The arm is sliding under the towel in a rigid environment. There is active sliding. No significant deformation is observed in the towel"
  }
}
```

Listing 2. Human-annotated labels for robot states.

```
{
  "state_1": {
    "code": "010-10001011-0000",
    "reason": "Quasi-static motion with potential energy, no grasp, sliding on a rigid environment with rigid contact, active sliding, and no deformation."
  },
  "state_2": {
    "code": "010-00111011-0000",
    "reason": "Quasi-static motion with potential energy, no grasp, sliding on a rigid environment with rigid contact, active sliding, and no deformation."
  }
}
```

Listing 3. VLM-annotated labels for robot states.